Module 13: Replication & High Availability

Overview

This module covers everything needed to design, implement, and operate a production-grade highly available PostgreSQL system. From the theory of WAL-based replication to complete Patroni cluster setups, HAProxy routing, and failover runbooks.

Prerequisites

- PostgreSQL administration basics (Modules 1-5)
- SQL proficiency
- Linux system administration
- Basic networking concepts

Module Contents

File	Topic	Key Concepts
01_replication_overview.md	Replication fundamentals	Physical vs logical, sync vs async
02_streaming_replication.md	Physical streaming replication	Full setup, pg_hba.conf, monitoring
03_logical_replication.md	Logical replication	Publications, subscriptions, migrations
04_synchronous_replication.md	Synchronous replication	synchronous_standby_names, quorum
05_failover_procedures.md	Failover and promotion	pg_ctl promote, pg_rewind, split-brain
06_patroni.md	Patroni HA automation	Architecture, patroni.yml, patronictl
07_pgbouncer.md	Connection pooling	Session/txn/statement modes, monitoring
08_haproxy_load_balancing.md	HAProxy load balancing	Health checks, read routing, Keepalived
09_ha_architecture_patterns.md	Complete HA architectures	ASCII diagrams, multi-region, capacity

Learning Path

```
Beginner:    01 → 02 → 03
Intermediate: 04 → 05 → 07
Advanced:     06 → 08 → 09
```

Key Architecture at a Glance

```
Apps --> HAProxy VIP --> PgBouncer --> PostgreSQL (Patroni)
                                            ↑
```

```
            etcd DCS cluster
                   ↓
           Automatic failover
```

Quick Reference: Which Replication Type?

Need	Use
HA hot standby	Physical streaming
Zero data loss	Synchronous physical
Selective table replication	Logical
Major version upgrade	Logical
Automatic failover	Patroni
Connection multiplexing	PgBouncer
Read/write routing	HAProxy

Related Modules

- Module 14: Security (access control for replication users)
- Module 15: Backup & Recovery (complements HA with point-in-time recovery)